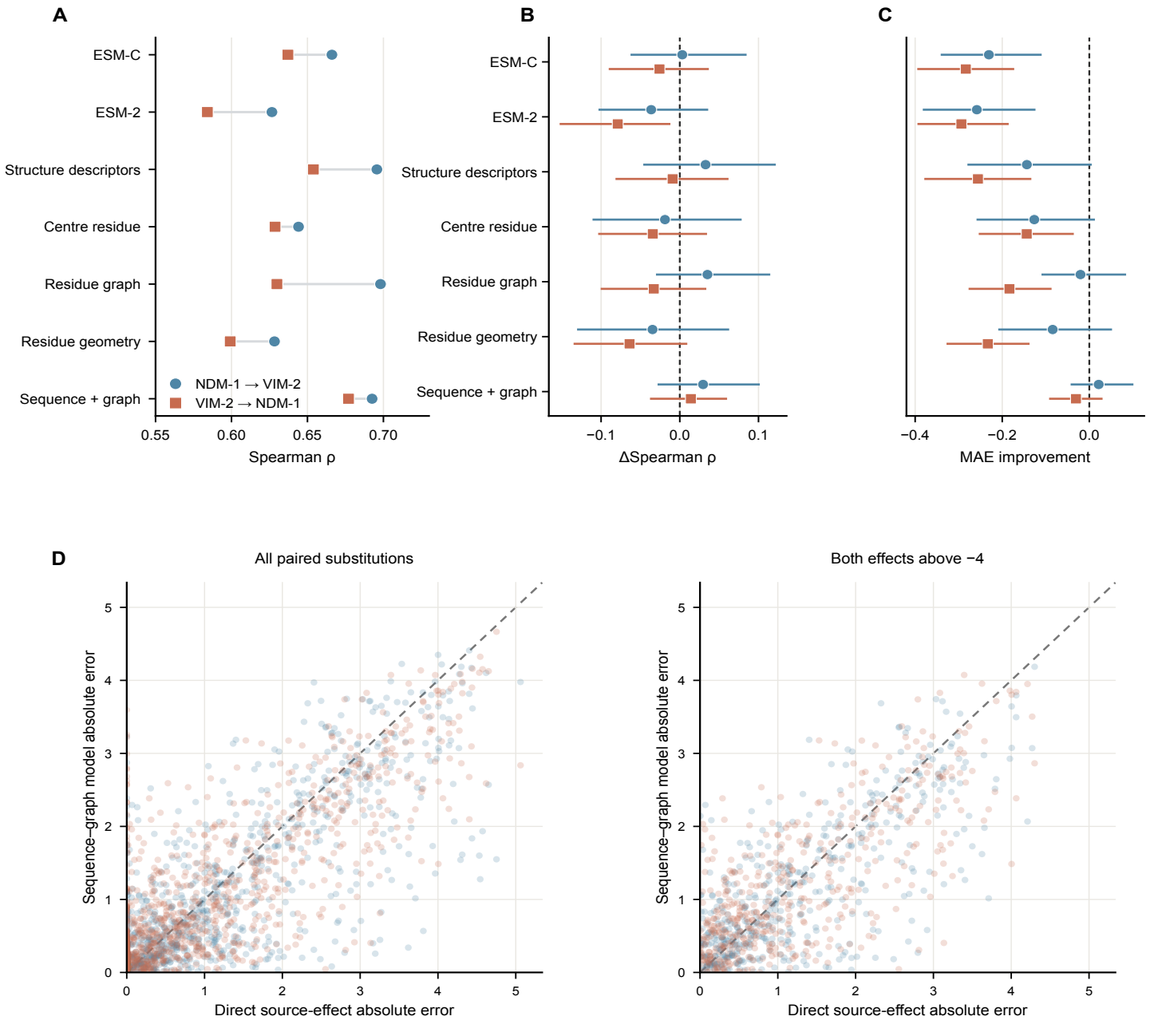

A
ESM-C
ESM-2
Structure descriptors
Centre residue
Residue graph
Residue geometry
Sequence + graph
NDM-1 → VIM-2
VIM-2 → NDM-1
0.55
0.60
0.65
0.70
Spearman ρ
B
ESM-C
ESM-2
Structure descriptors
Centre residue
Residue graph
Residue geometry
Sequence + graph
−0.1
0.0
0.1
ΔSpearman ρ
C
−0.4
−0.2
0.0
MAE improvement
D
All paired substitutions
Both effects above −4
Sequence–graph model absolute error
Direct source-effect absolute error
0
1
2
3
4
5